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## Summary

FSA with life and reinsurance experience across IFRS 17 reporting (IFRI Certificate in progress), experience studies, pricing support, and actuarial automation with strong ownership, documentation, and stakeholder communication.

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## Skills & Competency

- Programming: Python, VBA, R, SAS (Advanced Certificate)
- Data Platforms & ETL: SQL Server/DW/SSIS/Agent Jobs, Databricks/SparkSQL
- Analytical Modelling: Microsoft Excel, Power Query, Power Pivot, DAX, OLAP
- Reporting & Visualization: Power BI, Zebra BI, Tableau
- Development Tools: Streamlit, Git/GitHub, Visual Studio
- Actuarial Modelling: GGY AXIS

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## Work History

### Senior Actuarial Associate, Munich Re - Central Life Reserving, Toronto, ON, 09/2024 - 05/2026

- IFRS 17 Close & Output Controls — Coordinated multi-BU close process covering data delivery/RefCalc run setup/validation
- Data Pipeline — Automated approved Run ID extraction in Python and validated Databricks loads with Power BI and Spark SQL
- RefCalc Reconciliation — Reconciled/investigated RefCalc and Central System differences in IFRS17 results
- Data Validation Tool — Built Python validation engine with Object-Oriented Programming and Git controls and Streamlit-based UI
- Run Status Tracker — Built Power BI close-readiness dashboard flagging missing data, stale assumptions, and run-setting issues
- Adjustment Submission Template — Optimized workbook (ex. FX refresh/reconciliations/output) with Databricks & VBA
- Power BI Management Report — Designed ZebraBI graphs (ex. ISR/CSM roll-forward/expenses/NB vs. IF/LC transitions); tested PowerBI report features (ex. slicers/filters/drill-through/PPT automation)

### Actuarial Associate, Munich Re - Biometric Research, Toronto, ON, 06/2022 - 08/2024

- Mortality Data QA Framework — Mapped E2E data lineage; built a data dictionary and issue log; reconciled legacy vs. new system records; designed SQL QA checks with A/E impact thresholds to prioritize material issues
- UL/GUL NAAR Data Review — Analyzed patterns by duration and corrected distorted exposure with SQL & Excel
- US IL Mortality Experience Studies — Analyzed across key risk segments and through pandemic periods; evaluated A/E credibility; quantified excess mortality; supported pricing/valuation basis review
- LTC Transaction Support — Validated client GLM assumptions and GGY AXIS implementation to identify embedded conservatism
- Mortality Improvement Monitoring — Built a flexible MI monitoring framework using Power Query/Power Pivot/DAX/parameter tables; enabled rapid recalculation for different views; enhanced assumption-governance documentation
- GLTD Study Leadership — Led SQL Server ETL and OLAP cube generation process with modular checkpoints/data quality controls while supervising Co-OP; presented results/analysis/recommendation to stakeholders

### Senior Actuarial/Consulting Analyst, Mercer - Health & Benefits, Ottawa, ON, 07/2021 - 05/2022

- Group Benefits Renewal Pricing — Reconciled premiums/claims/reserves/fees; analyzed utilization/severity/demographics/benefit mix.
- Credibility & Funding Strategy — Assessed experience credibility and evaluated funding/reserves/pooling/SL/contribution strategies
- Claims & Financial Audit — Audited claims/financials against plan design/agreements, identified discrepancies & recommended remediation

### Actuarial Assistant, Reinsurance Group of America - Global Risk Service, Toronto, ON, 09/2020-12/2020

- Economic Capital/Risk Limit Model — Maintained and refreshed Excel/VBA-based model to calculate stand-alone capital etc; performed sensitivity testing to assess risk limit utilization
- KRI Dashboard — Built Excel/VBA dashboard for biometric/lapse/recapture/downgrade/RBC trigger/treaty termination risks
- Quarterly Treaty Data Reconciliation — Investigated and corrected recapture and termination GAAP loss exposure in Excel
- GGY AXIS Training Materials — Updated to reflect software interface updates and Corporate Pricing Manual changes

### Actuarial Assistant, Reinsurance Group of America - Global Product, Toronto, ON, 09/2019-12/2019

- Retrocession Premium Audit — Developed an Excel/VBA audit tool to independently recalculate retrocession premiums at the policy level, reflecting treaty-specific rating structures/mortality assumptions/policy provisions/joint-life calculation logic; documented root causes of discrepancies and presented recommendations internationally.
- Product Innovation Research — Researched customer segmentation and product trends for RGA's global actuarial seminar

### Actuarial Student, Canada Life - Guaranteed Product & Pricing, London, ON, 01/2019-04/2019

- Annuity Special Quotes — Prepared larger-benefit and older-age annuity quotes in UI-based Annuity Quoting System and tested using Excel cash flow models with updated yield curves to assess longevity/concentration risk and ROI
- LICAT Annuity Cashflow Model Update — Updated the Excel/VBA model to align with LICAT mortality risk requirements for life-supported benefits (ex. CSV/RoP); fixed a backdated policy issue affecting calculation involving policy age/effective dates/mortality table application during accumulation period

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## Education

**Bachelor of Honours Mathematics: Actuarial Science & Statistics**  
University of Waterloo, Waterloo, ON

April 2021

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